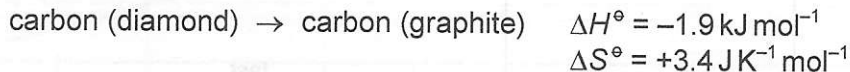


10 Use of the Data Booklet is relevant to this question.

Carbon can exist naturally in two different crystalline forms, diamond and graphite.



Which statements are correct **and** explain why diamond does **not** change into graphite at room temperature and pressure?

- 1 The forward reaction has an extremely small rate constant.
- 2 ΔG° is -2.9 kJ mol^{-1} for the above transformation.
- 3 The reverse reaction is spontaneous.

A 1 and 3

B 1 only

C 2 and 3

D 2 only